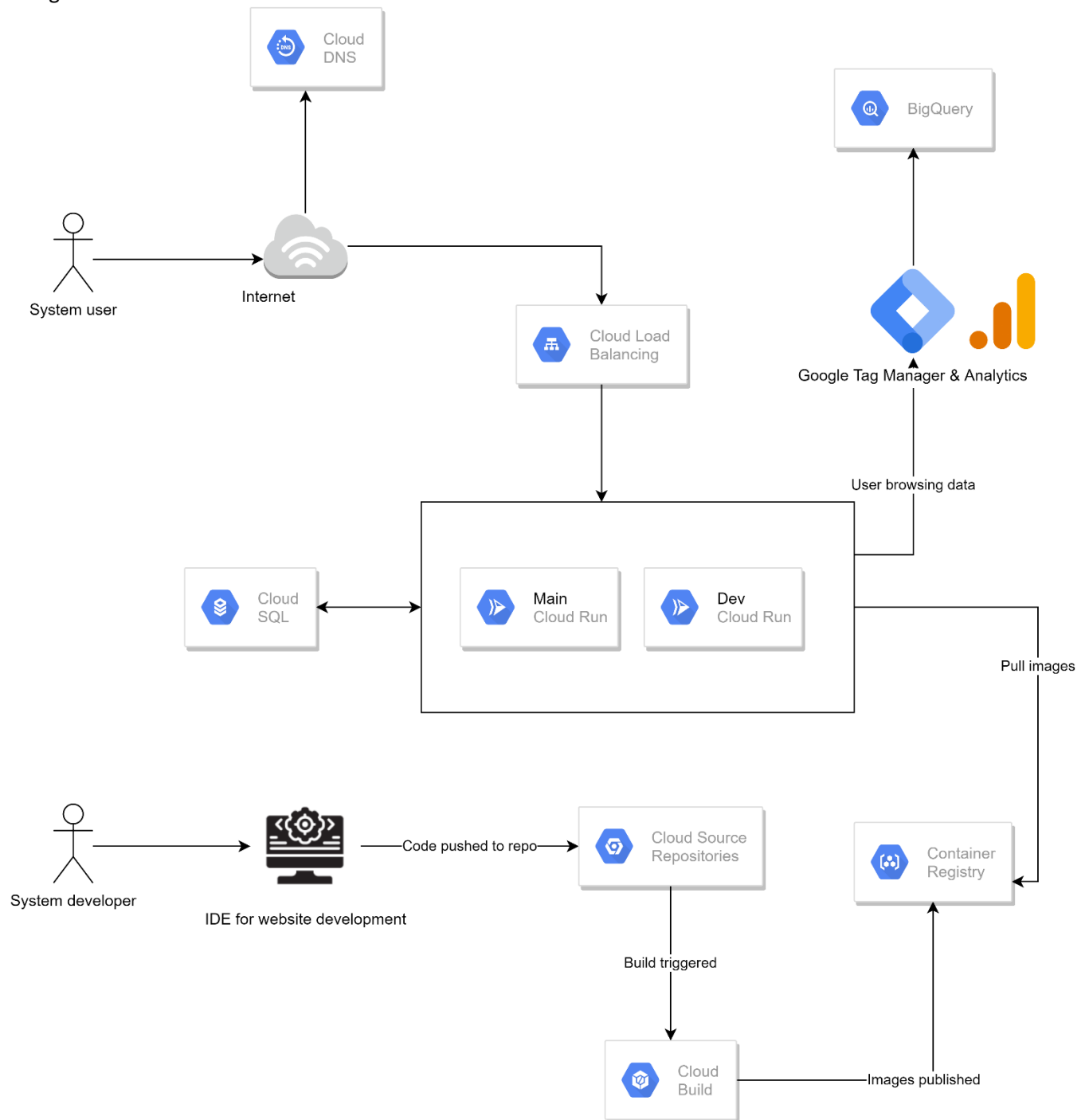


Cloud-based system redesigned on another service provider platform

As before, the system is a landing-page type website that informs the user of the features and details of an upcoming project. The purpose is to present people a soon-to-be-released project, its release date, feature list, and to attract as many potential new users as possible. The system has been redesigned for Google Cloud Platform.



The proposed altered architecture would use Google Cloud Platform services. This time, from the development side, it is also assumed that a single web developer is working on the project using an IDE of their choice, however this time they push or pull code that is on Google Cloud Source Repositories, assuming that the website once again has one repository with two branches – dev and main. Once new code is pushed to either branch, a build is triggered for the respective branch to Cloud Build service, and an image is published to Google Container Registry. Google Cloud Run services are used for web app deployment (main and dev environments), and images for respective versions of the website are pulled from Container Registry.

Once again from the system user side, the website can be accessed via the Internet using a device of their choice (computer, a tablet, or smartphone). The website can be visited by entering a domain provided by the Google Cloud DNS service. For two Cloud Run service web apps (one for development purposes, and the main one), two DNS can be set up, for example dev.somewebsite.com and somewebsite.com. To prepare for a possibly large number of users and maintain the availability of the website, the Google Cloud Load Balancing service would be used to distribute the load evenly. Google Cloud SQL service may be used for the database, with at least two tables (one for storing user email information for sending newsletters and one for other miscellaneous information dynamically displayed on the website). The data flow is once again two-way, as the website may not only write information to the database but also pull information from it. This time, user browsing data (such as data of buttons clicked, the amount of time spent on the website, the amount of successful newsletter subscriptions, etc.) is managed by external Google services – Google Tag Manager and Google Analytics. Google Tag Manager is used to set up events that are executed whenever a tag is fired, e.g., a website is visited or a button is clicked, and the data is displayed on Google Analytics. This time, instead of Azure Storage Blob, the Google BigQuery service can be used to store large amounts of data via a one-way data flow from Google Analytics.